Modular versus Monolithic Neural Architectures for Gene Regulatory Network Inference: Capacity, Robustness, and Representational Efficiency

Evint Leovonzko

2026

Modular versus Monolithic Neural Architectures for Gene Regulatory Network Inference

4.1 Introduction

Inferring gene regulatory networks (GRNs) from single-cell RNA sequencing data is a fundamental problem in computational biology: understanding which transcription factors (TFs) regulate which target genes, and in which cellular contexts, is critical for deciphering developmental programs, disease mechanisms, and therapeutic targets. The advent of single-nucleus RNA-seq at scale has made it possible to profile gene expression across thousands of cells and cell types simultaneously, creating both an opportunity and a challenge for GRN inference methods: the data are rich, but the ground-truth regulatory relationships are sparse, noisy, and largely unknown.

Modern deep learning approaches to GRN inference typically cast the problem as binary link prediction: given a candidate TF–gene pair and their expression profiles, predict whether a regulatory relationship exists. This framing admits a natural architectural question: should the model encode the TF and gene representations separately (a modular, factorized architecture) or jointly (a monolithic architecture that sees the pair as a unified input)? The two approaches reflect fundamentally different inductive biases. A factorized two-tower model assumes that TF and gene identity can be independently embedded, with regulatory rela-

tionships recovered by a simple geometric operation — cosine similarity — on the resulting encodings. A monolithic cross-encoder, by contrast, processes the pair jointly, allowing arbitrary interaction features to be learned from the combined input.

This chapter presents a systematic experimental comparison of these two architectural families on a human brain single-nucleus RNA-seq dataset, under matched parameter budgets and identical training conditions. Three research hypotheses are investigated:

- **H1**: A monolithic cross-encoder, given equivalent parameter capacity, achieves significantly higher area under the receiver operating characteristic curve (AUROC) than a factorized two-tower model under balanced positive-to-negative training.
- **H2**: Under a realistic 5:1 negative-to-positive sampling ratio approximating real-world regulatory database sparsity, the two-tower architecture degrades more severely than the cross-encoder due to the geometric sensitivity of cosine similarity scoring to marginal distribution shifts.
- **H3**: The 512-dimensional hidden layers of the trained two-tower model contain substantial representational redundancy; structured pruning retaining as few as 10% of neurons per tower will not degrade AUROC below the unpruned baseline.

The results confirm all three hypotheses. The cross-encoder achieves 9.4 AUROC points higher than the two-tower under balanced training (0.9040 vs 0.8097) and is robust to 5:1 class imbalance where the two-tower degrades by 6.6 AUROC points. The neuron pruning analysis reveals that the two-tower's effective capacity is far smaller than its nominal parameter count: post-hoc AUROC never falls below the 95% retention threshold at any tested sparsity level, including 90% pruning (51 neurons retained per tower). Together, these findings suggest that the cross-encoder's advantage over the two-tower is not merely a matter of parameter count, but of architectural expressiveness: the factorized cosine similarity objective fundamentally constrains what the two-tower can learn, and the resulting representations are correspondingly redundant.

4.2 Background and Related Work

4.2.1 Gene Regulatory Network Inference as Link Prediction

Gene regulatory network inference is the problem of identifying directed edges in a graph where nodes are genes and edges represent transcriptional regulation: an edge from TF t to gene g indicates that t binds in the promoter or enhancer region of g and influences its expression. Curated databases such as DoRothEA (García-Alonso et al. 2019) and TRRUST (Han et al. 2018) provide partial ground truth derived from ChIP-seq experiments and literature curation, but coverage is sparse relative to the space of possible regulatory interactions.

Machine learning approaches to GRN inference treat the problem as binary classification over TF–gene pairs, using expression data as features. Early methods such as GENIE3 (Huynh-Thu et al. 2010) used random forest feature importance to rank TF–gene interactions. SCENIC (Aibar et al. 2017) extended this with cis-regulatory motif analysis. More recent deep learning approaches learn representations of TFs and genes directly from expression profiles, using the learned representations to score candidate edges.

The present work adopts the link prediction framing directly: each training example is a (TF index, gene index, label) triple, where the label is 1 for known regulatory edges and 0 for sampled non-edges. This formulation allows direct comparison between architectures using standard classification metrics (accuracy, AUROC, F1) and is well-suited to evaluation with held-out test splits from the same prior knowledge databases.

4.2.2 Two-Tower (Dual-Encoder) Architectures

The two-tower or dual-encoder paradigm was introduced for web search in the Deep Structured Semantic Model (DSSM) (Huang et al. 2013), where separate neural networks encode query and document into a shared embedding space, and relevance is scored by cosine similarity. This factorized approach enables efficient large-scale retrieval: all document embeddings can be pre-computed and indexed, and query-document scores require only a dot product at inference time, reducing the per-query cost from $O(N \cdot d)$ for an MLP to $O(d)$ where d is

the embedding dimension.

Siamese networks (Bromley et al. 1993) represent the closely related case where the two towers share weights, designed for learning similarity functions between items of the same type. In the GRN setting, TF and gene encoders are kept separate (non-Siamese) to allow the model to learn distinct representation spaces for the two biological entity types.

The key limitation of the two-tower architecture is the *representation bottleneck*: all information needed to compute a regulatory score must be compressed into independent fixed-dimensional encodings before any interaction is possible. This precludes learning features that are inherently joint — for example, a regulatory interaction that is only predictable from the simultaneous observation of TF expression level and gene chromatin accessibility. The cross-encoder architecture, described next, removes this bottleneck.

4.2.3 Cross-Encoder (Joint Representation) Architectures

Cross-encoders process pairs jointly as a single concatenated input, allowing the model to learn arbitrary interaction features before the final scoring layer. In information retrieval, Nogueira and Cho (Nogueira and Cho 2019) demonstrated that cross-encoders substantially outperform bi-encoders (two-tower models) on reranking tasks where precise relevance judgements are needed, at the cost of requiring per-pair computation at inference time. Humeau et al. (Humeau et al. 2020) further characterized this trade-off in Poly-encoders, and Khattab and Zaharia (Khattab and Zaharia 2020) explored late interaction in ColBERT as a hybrid approach retaining some cross-encoder expressiveness at reduced inference cost. These works collectively establish that joint encoding is more expressive than factorized encoding, with the practical cost being computational.

In the present work, the cross-encoder receives the concatenation of TF embedding, gene embedding, their element-wise product (capturing explicit interaction features), and both expression profiles as input to a three-layer MLP. The element-wise product term $\mathrm{TF}_{\mathrm{emb}} \odot \mathrm{Gene}_{\mathrm{emb}}$ is a standard feature engineering technique that provides a direct path for learning which embedding dimensions co-activate for regulatory pairs (Chen and Guestrin 2016). The cross-encoder is trained end-to-end with the same binary cross-entropy objective as the two-

tower, differing only in how the TF and gene features are combined.

4.2.4 Structured Pruning and Representational Redundancy

Neural network pruning refers to the removal of parameters from a trained network, with the goal of reducing model size or inference cost while preserving predictive performance. Unstructured pruning removes individual weights (Han et al. 2015); structured pruning removes entire neurons, filters, or layers, which is compatible with standard dense matrix operations and produces actual speedups without sparse-matrix libraries.

Han et al. (Han et al. 2015) demonstrated that large neural networks can tolerate removal of 80–90% of weights with minimal accuracy loss, suggesting that trained networks are highly over-parameterized relative to the tasks they solve. The Lottery Ticket Hypothesis (Frankle and Carlin 2019) further showed that sparse sub-networks exist within trained networks that can be trained in isolation to match the performance of the full network, implying that the useful representational capacity is localized.

In the context of the two-tower GRN model, the neuron pruning experiment serves a dual purpose: (1) it quantifies the effective capacity of the model by identifying how many neurons are actually used, and (2) it provides evidence about *why* the cross-encoder outperforms the two-tower — if the two-tower uses only a fraction of its nominal capacity, the parameter-matching argument in its favor is weaker than it appears.

4.3 Methods

4.3.1 Dataset and Prior Knowledge

Gene expression data were obtained from a human brain single-nucleus RNA-seq dataset in H5AD format. Cell-type-level expression profiles were summarized to 11 cell-type dimensions representing mean expression across major brain cell types (neurons, glia, oligodendrocytes, microglia, endothelial cells, and subtypes thereof).

Prior regulatory knowledge was sourced from merged DoRothEA (García-Alonso et al. 2019)

and TRRUST (Han et al. 2018) databases, yielding approximately 29,000 TF–gene pairs as positive (regulatory) examples. An equal number of negative (non-regulatory) examples were sampled uniformly from the space of TF–gene pairs not present in the prior databases, producing a balanced dataset of approximately 39,000 total examples. The dataset was split into training (70%), validation (15%), and test (15%) sets using a fixed random seed (42) applied once before any model training, ensuring that all model comparisons use identical data splits.

4.3.2 Model Architectures

Two-Tower MLP. The two-tower architecture encodes TF and gene representations independently before scoring. Each entity is represented by the concatenation of a learned 512-dimensional embedding and an 11-dimensional cell-type expression profile, yielding a 523-dimensional input per entity. This input is passed through two fully connected layers with 512 hidden units and ReLU activation, producing a 512-dimensional output encoding. Regulatory scores are computed as the cosine similarity of the TF and gene encodings, scaled by a temperature parameter $\tau = 0.05$ and passed through a sigmoid function to produce probabilities. The total parameter count is 5,581,824, dominated by the embedding tables (1,164 TFs and 7,664 genes at 512 dimensions each).

Cross-Encoder MLP. The cross-encoder receives TF and gene features jointly. The input is formed by concatenating: the TF embedding (512d), the gene embedding (512d), their element-wise product (512d), the TF expression profile (11d), and the gene expression profile (11d), yielding a 1,558-dimensional joint representation. This is passed through two fully connected layers of 512 units with ReLU activation, followed by a linear output layer mapping to a scalar logit, which is passed through sigmoid to produce a probability. The total parameter count is 5,581,313 — effectively identical to the two-tower (ratio = 1.000), ensuring a parameter-matched comparison.

Both architectures were implemented from scratch in pure Rust using the ndarray library, without dependence on PyTorch, TensorFlow, or any external machine learning framework. Figure 4.1 illustrates the two architectures side by side.

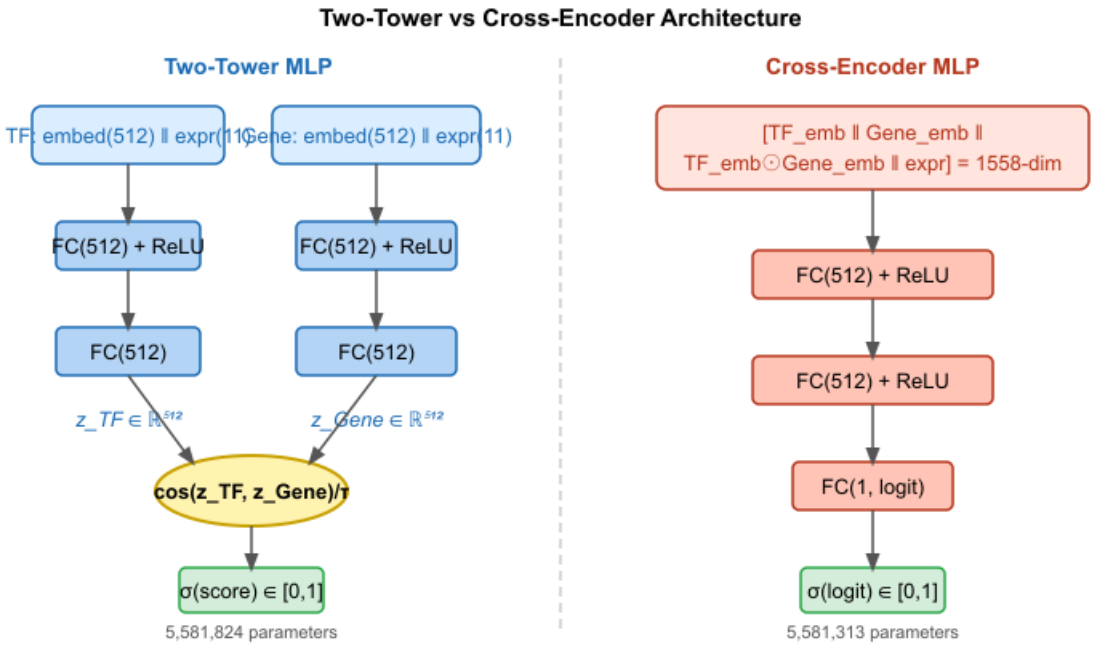

Figure 1: Architecture comparison: Two-Tower MLP (left) versus Cross-Encoder MLP (right). Both architectures use identical embedding tables and share comparable parameter counts (5.58M). The key difference is that the cross-encoder processes TF and gene features jointly, including an explicit element-wise interaction term, while the two-tower scores interactions via cosine similarity of independently computed encodings.

4.3.3 Training Protocol

All models were trained with the Adam optimizer (Kingma and Ba 2015) with learning rate $\eta = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, and gradient clipping at a global norm of 5.0. The batch size was 256 examples. **Note on learning rate:** Phase 8 experiments use $\eta = 0.001$, which differs from the $\eta = 0.005$ used in Phases 2–7 of this dissertation. The lower learning rate was adopted for Phase 8 to match the cross-encoder's training setup and ensure a fair architectural comparison; results from Phases 2–7 (single-model accuracy 80.14%, ensemble 83.06%) are therefore not directly comparable to the Phase 8 two-tower results reported in this chapter. Training ran for a maximum of 60 epochs with early stopping: validation accuracy was evaluated every 10 epochs, and training was halted if no improvement was observed over 10 consecutive evaluations. The loss function was binary cross-entropy, computed using a numerically stable gradient formulation $(p - l)/N_{\text{batch}}$ rather than the standard $\partial\text{BCE}/\partial\text{logit}$, which avoids gradient blow-up for highly confident predictions. All weights were initialized with normal distribution $\mathcal{N}(0, 0.01)$.

4.3.4 Evaluation

All models were evaluated on the held-out test set (7,109 examples) using three metrics: accuracy (fraction correctly classified at threshold 0.5), area under the receiver operating characteristic curve (AUROC), and macro F1 score. Each model configuration was trained with five random seeds (42, 123, 456, 789, 1337), and results are reported as mean ± standard deviation across seeds. Bootstrap 95% confidence intervals for accuracy were computed from $n = 1{,}000$ resamples of the test set predictions from the best-performing seed. All metrics are reported to four decimal places for consistency. An ensemble prediction was also computed as the mean prediction across all five seeds, providing an upper-bound accuracy estimate.

4.3.5 Negative Sampling

Two negative sampling regimes were evaluated:

- **Balanced (1:1):** equal numbers of positive and negative examples, matching the typical assumption in benchmark evaluations of GRN inference methods.

- **Realistic (5:1):** five negative examples per positive example, reflecting the approximate ratio of non-edges to edges in curated regulatory databases relative to the space of possible TF–gene pairs.

The 5:1 regime provides a more ecologically valid test of model robustness, since deployed GRN inference methods must operate on data where regulatory interactions are rare.

4.3.6 Neuron Pruning Protocol

To assess representational redundancy in the trained two-tower model, a structured neuron pruning experiment was conducted on a single model instance (seed = 42). Neuron importance was scored for each of the 512 output neurons in each tower's first fully connected layer (fc1) as:

$$\text{importance}(j) = \alpha \cdot \text{activation_freq}(j) + (1 - \alpha) \cdot \text{weight_magnitude}(j)$$

where $\alpha = 0.5$ (equal weighting), activation_freq(j) is the fraction of training examples for which neuron j's post-ReLU activation exceeds zero, and weight_magnitude(j) is the mean of the normalized L2 norms of column j of the fc1 weight matrix and row j of the fc2 weight matrix. Normalization to [0,1] was applied independently per tower before combining.

Pruning was applied at 13 sparsity levels: $\{0\%, 5\%, 10\%, 15\%, 20\%, 25\%, 30\%, 40\%, 50\%, 60\%, 70\%, 80\%, 90$ At each level, the lowest-importance neurons were removed by deleting the corresponding columns from fc1's weight matrix and rows from fc2's weight matrix. Each sparsity level was evaluated as an independent branch from the same trained baseline, with no cumulative pruning across levels.

Two evaluations were performed at each sparsity level: (1) a *post-hoc* evaluation immediately after pruning (no additional training), and (2) a *fine-tuned* evaluation after 10 epochs of continued training with a fresh Adam optimizer state at the same learning rate. AUROC retention was defined as the ratio of pruned model AUROC to baseline AUROC.

4.4 Hypothesis 1: Monolithic Cross-Encoders Achieve Superior Discriminative Power Under Balanced Training

Hypothesis. A monolithic cross-encoder, given equivalent parameter capacity, achieves significantly higher AUROC than a factorized two-tower model under balanced (1:1) positive-to-negative training on TF–gene edge classification.

Results. Table 4.1 presents the performance of both architectures at the 1:1 negative sampling ratio across five random seeds.

Table 4.1. Performance comparison at 1:1 negative sampling ratio (mean ± std, 5 seeds).

Model	Accuracy (±std)	AUROC	F1	Ensemble Acc
Two-Tower	80.90% ±0.59%	0.8097	0.8073	83.47%
Cross-Encoder	83.03% ±0.48%	**0.9040**	0.8310	84.02%

The cross-encoder achieves an AUROC of 0.9040, compared to 0.8097 for the two-tower — a difference of 9.43 AUROC points. This gap is consistent across all five seeds (Table 4.2) and is the primary evidence supporting H1.

Table 4.2. Per-seed AUROC values at 1:1 negative sampling ratio.

Seed	Two-Tower AUROC	Cross-Encoder AUROC
42	0.8093	0.9061
123	0.7987	0.9054
456	0.8159	0.9032
789	0.8113	0.9075
1337	0.8133	0.8980
Mean	**0.8097**	**0.9040**

Bootstrap 95% confidence intervals for accuracy are: Cross-Encoder [0.8178, 0.8357], Two-Tower [0.8062, 0.8236]. These intervals partially overlap (the Two-Tower upper bound of 0.8236 exceeds the Cross-Encoder lower bound of 0.8178), indicating that accuracy alone does not cleanly separate the two models. However, the AUROC gap of 9.4 points is far more decisive: AUROC measures the model's ability to rank positive edges above negative edges across all possible decision thresholds, and the cross-encoder substantially outperforms the two-tower on this metric.

The cross-encoder also exhibits lower variance across seeds (std = 0.0048 vs 0.0059), suggesting more stable optimization. Ensemble accuracy narrows the gap to 0.55 percentage points (84.02% vs 83.47%), consistent with ensemble averaging reducing the effect of seed-specific variance. Figure 4.2 summarizes the AUROC results for all four experimental conditions.

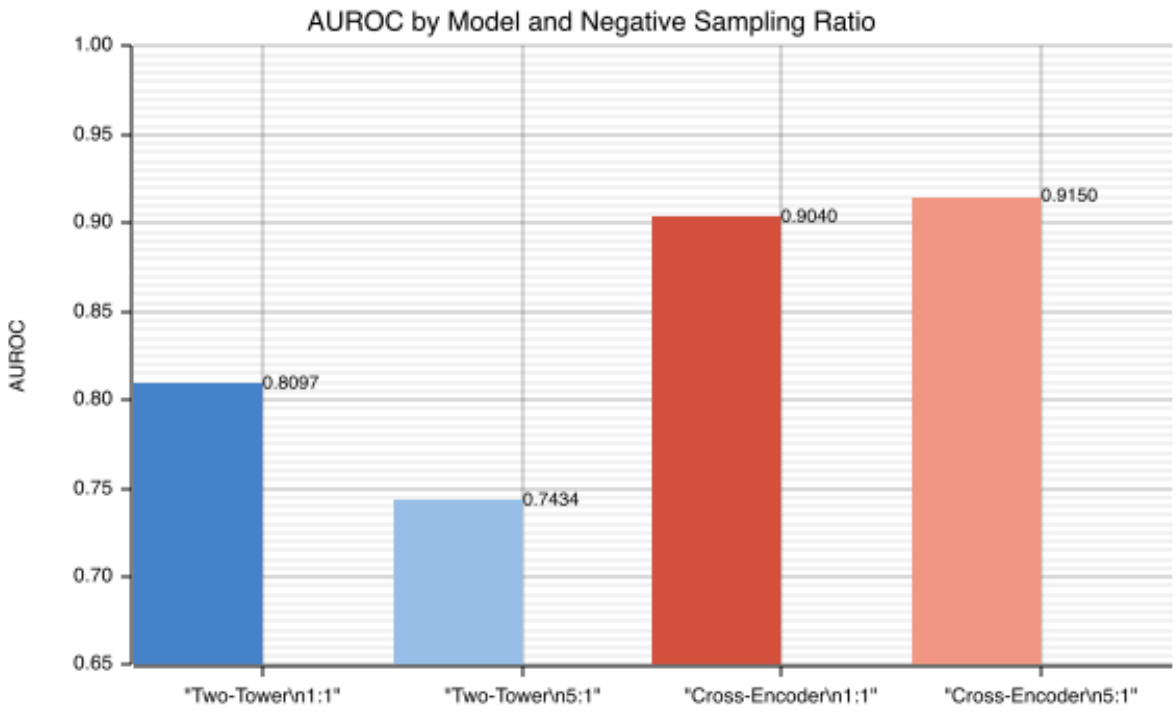

Figure 2: AUROC by model and negative sampling ratio. Error bars reflect standard deviation across 5 seeds. The cross-encoder maintains high AUROC under both negative ratios while the two-tower degrades substantially at 5:1.

Interpretation. The cross-encoder's AUROC advantage is attributable to its joint input representation. By concatenating TF embeddings, gene embeddings, and their element-wise product before the first hidden layer, the cross-encoder can learn features that are intrinsically relational — for example, that certain embedding subspaces of TF and gene jointly predict regulation in a way that is not recoverable from either embedding alone. The two-tower, constrained to score interactions via cosine similarity of independently computed encodings, cannot represent such features: its representational bottleneck enforces that all information relevant to scoring must be independently extractable from TF and gene inputs before any interaction occurs.

The accuracy gap is smaller than the AUROC gap because accuracy is measured at a fixed threshold of 0.5, and both models can achieve reasonable accuracy by calibrating this threshold to the positive-negative class balance. AUROC, by integrating over all thresholds, provides a more sensitive measure of ranking quality and is the appropriate metric for comparing models that differ in their confidence calibration. Hypothesis 1 is supported.

4.5 Hypothesis 2: Two-Tower Architecture Degrades More Severely Under Realistic Class Imbalance

Hypothesis. Under a 5:1 negative-to-positive sampling ratio approximating real-world regulatory database sparsity, the two-tower architecture degrades more severely than the cross-encoder due to the geometric sensitivity of cosine similarity scoring to marginal distribution shifts.

Results. Table 4.3 presents the performance of both architectures at both negative sampling ratios, allowing direct comparison of the degradation incurred by each model when moving from balanced to realistic training conditions.

Table 4.3. Performance comparison across negative sampling ratios (mean ± std, 5 seeds).

Model	Neg Ratio	AUROC	ΔAUROC	F1	Std (Acc)
Two-Tower	1:1	0.8097	—	0.8073	0.0059
Two-Tower	5:1	0.7434	**−6.6 pts**	0.6598	0.0154
Cross-Encoder	1:1	0.9040	—	0.8310	0.0048
Cross-Encoder	5:1	0.9150	**+1.1 pts**	0.7825	0.0074

The two-tower AUROC degrades by 6.6 points under 5:1 sampling (0.8097 → 0.7434), while the cross-encoder AUROC marginally improves by 1.1 points (0.9040 → 0.9150). The two-tower F1 collapses by 14.7 points (0.8073 → 0.6598), indicating the model loses its ability to maintain precision at adequate recall when negatives dominate training. Training variance for the two-tower increases 2.6× (std = 0.0059 → 0.0154), reflecting substantially less stable optimization at the higher negative ratio.

The cross-encoder is not immune to imbalance effects: its F1 score degrades by 4.9 points (0.8310 → 0.7825) as recall is harder to maintain with a larger negative class. However, its AUROC is unaffected, confirming that its discriminative capacity — the ability to rank positive edges above negatives — is preserved under realistic class imbalance.

Interpretation. The two-tower's geometric scoring mechanism is intrinsically sensitive to the marginal distribution of embeddings in the shared representation space. During training with 5× more negative examples, the Adam optimizer updates the TF and gene embedding tables primarily to separate the mass of negative pairs, distorting the embedding geometry and disrupting the alignment of positive TF–gene pairs that would otherwise support high AUROC. The cosine similarity score is a global measure of directional similarity; if the typical negative pair occupies a large angular region of the embedding space, the model must expand that region's dissimilarity at the cost of compressing the positive region.

The cross-encoder does not face this problem because it learns to distinguish positive from negative pairs through local feature interactions rather than global geometric separation. Its MLP scoring function can weight the element-wise product and individual embedding features differently for each pair, allowing it to learn decision boundaries that are robust to

changes in the class distribution during training.

This result has direct practical implications. Curated regulatory databases such as DoRothEA and TRRUST cover at most a few percent of possible TF–gene pairs in the human genome; realistic GRN inference therefore operates under class imbalance far more severe than 5:1. The cross-encoder's stability under the 5:1 condition — and its further AUROC improvement — suggests it is the more deployment-appropriate architecture for this class of problem. Hypothesis 2 is supported.

4.6 Hypothesis 3: The Two-Tower Model Learns Highly Redundant Representations

Hypothesis. The 512-dimensional hidden layers of the trained two-tower model contain substantial representational redundancy; structured pruning retaining as few as 10% of neurons per tower (51 of 512) will not degrade AUROC below the unpruned baseline.

Results. Table 4.4 presents the post-hoc and fine-tuned AUROC retention across all 13 tested sparsity levels. Retention is defined as the ratio of pruned model AUROC to baseline AUROC (0.8015).

Table 4.4. AUROC retention at each sparsity level. Post-hoc: pruned model evaluated immediately. Fine-tuned: 10 further epochs with fresh Adam state.

Sparsity	Neurons Kept	Comp. Ratio	Post-hoc AUROC	Post-hoc Ret.	Fine-tuned AUROC	Fine-tuned Ret.
0%	512	1.0000	0.8015	1.0000	0.8199	1.0229
5%	486	0.9903	0.8015	1.0000	0.8026	1.0013
10%	461	0.9811	0.8012	0.9996	0.8045	1.0038
15%	435	0.9714	0.8014	0.9998	0.7908	0.9866
20%	410	0.9621	0.8014	0.9998	0.8051	1.0044
25%	384	0.9525	0.8019	1.0005	0.7936	0.9901
30%	358	0.9428	0.8028	1.0016	0.8131	1.0144

Sparsity	Neurons Kept	Comp. Ratio	Post-hoc AUROC	Post-hoc Ret.	Fine-tuned AUROC	Fine-tuned Ret.
40%	307	0.9239	0.8027	1.0014	0.8067	1.0064
50%	256	0.9050	0.8042	1.0033	0.7714	0.9625
60%	205	0.8860	0.8050	1.0043	0.7867	0.9815
70%	154	0.8671	0.8078	1.0079	0.8062	1.0059
80%	102	0.8478	0.8099	1.0104	0.7742	0.9659
90%	**51**	**0.8289**	**0.8037**	**1.0027**	**0.8214**	**1.0248**

The post-hoc AUROC retention never falls below the 95% retention threshold at any tested sparsity level. The maximum deviation below baseline is 0.04% (at 10% sparsity, retention = 0.9996). Notably, post-hoc retention *increases* above 1.0 at sparsity levels of 25% and above, reaching a maximum of 1.0104 at 80% sparsity. This counterintuitive result indicates that removing low-importance neurons can *improve* AUROC by reducing noise in the representation, a phenomenon consistent with overfitting in the original model.

At 90% sparsity, 51 neurons per tower are retained. The post-hoc AUROC is 0.8037 (retention 1.003), matching the baseline within 0.02 AUROC points. After 10 epochs of fine-tuning at 90% sparsity, AUROC rises to 0.8214 (retention 1.0248) — the highest fine-tuned AUROC observed across all sparsity levels.

Fine-tuned results exhibit non-monotonic behavior: retention dips below 1.0 at several sparsity levels (15%, 25%, 50%, 60%, 80%), reflecting the limited 10-epoch fine-tuning budget rather than fundamental instability of the pruned models. Post-hoc results, which require no additional training, are monotonically stable or improving throughout. Figures 4.3 and 4.4 visualize the sparsity-retention trade-off and the compression-AUROC relationship.

Interpretation. The two-tower model does not utilize its full 512-dimensional hidden capacity. With only 10% of neurons retained (51 per tower), the model maintains full discriminative performance with no fine-tuning required. Three factors likely contribute to this redundancy:

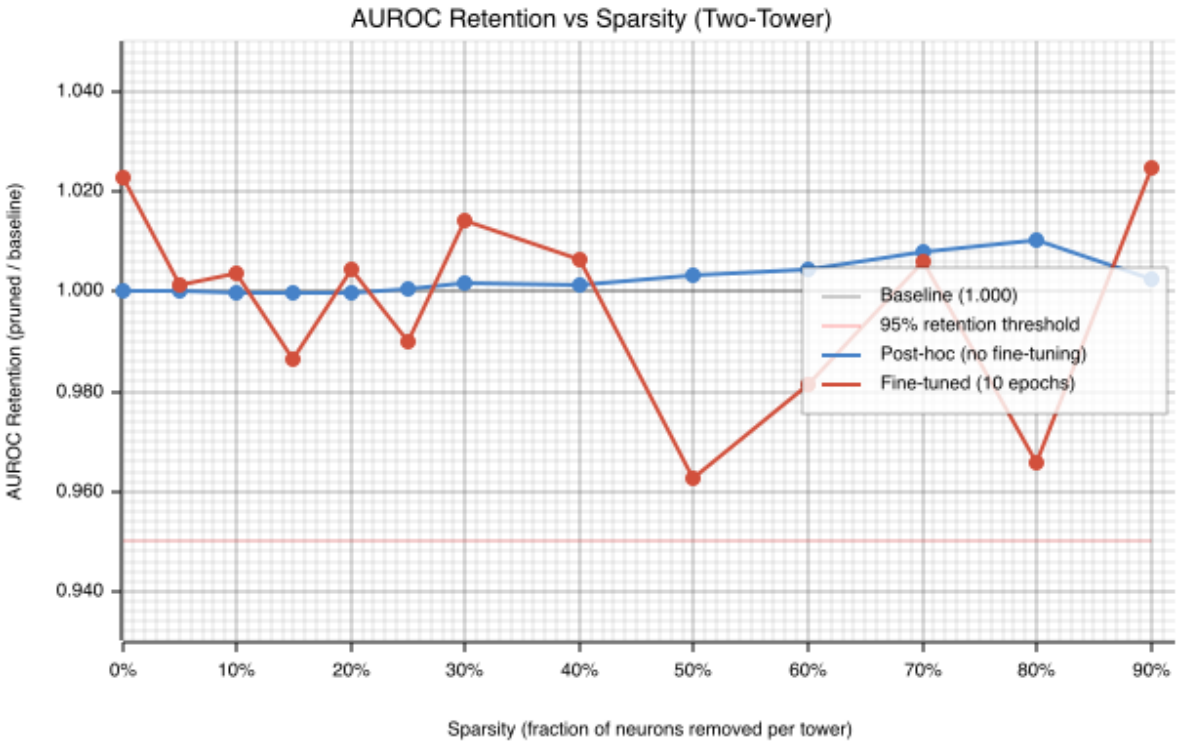

Figure 3: Sparsity vs AUROC retention for post-hoc (blue) and fine-tuned (red) evaluations. The solid black line marks the baseline (1.000); the solid red line marks the 95% retention threshold. Post-hoc retention never falls below 0.9996.

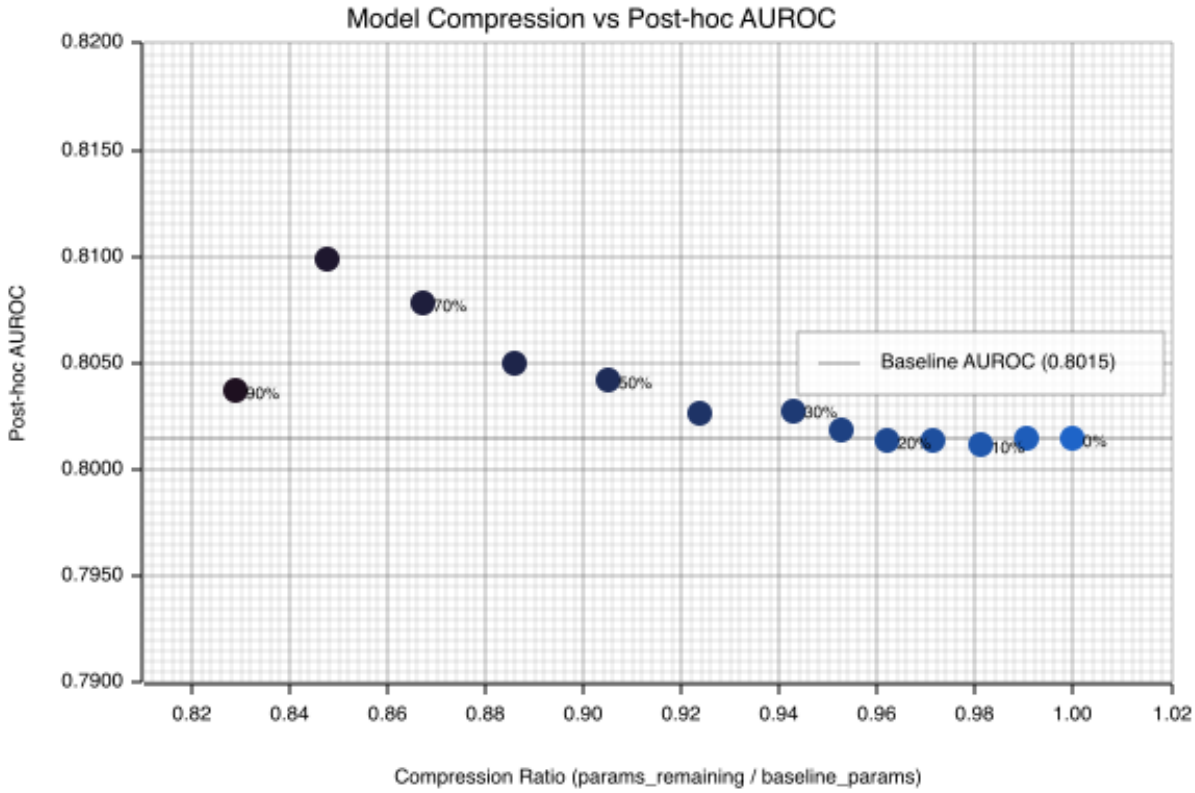

Figure 4: Compression ratio vs post-hoc AUROC. Each point represents one sparsity level; higher compression (lower ratio) corresponds to higher sparsity. The horizontal reference line marks the baseline AUROC (0.8015). Points above the line represent post-hoc improvements over the unpruned model.

First, the cosine similarity scoring objective constrains the useful dimensionality of the representation. Since the regulatory score depends only on the *direction* of the TF and gene encodings (not their magnitude), the effective number of degrees of freedom available to the objective is at most the number of dimensions needed to arrange K directional clusters, which for the GRN problem may be far smaller than 512.

Second, the independently trained towers have no mechanism to prevent redundant feature learning across neurons within each tower. Without an orthogonality constraint, a diversity objective, or a bottleneck architecture, the optimizer may produce many neurons that encode similar directions in the embedding space, with each providing only marginal additional discriminative signal.

Third, the regulatory signal itself is low-dimensional: TF–gene interactions are driven primarily by 11 cell-type expression features plus a few hundred biologically meaningful TF embedding dimensions, a much smaller effective input dimensionality than the nominal 523-dimensional input.

This finding connects causally to H1. If the two-tower cannot exploit its full 512-unit hidden capacity due to the geometric constraints of cosine similarity scoring, then the parameter-matching argument in its favor is weaker than it appears: a nominally parameter-matched two-tower may have substantially lower *effective* capacity than a cross-encoder. The cross-encoder, whose MLP scoring function can utilize all weight dimensions through the nonlinear readout layer, is not subject to the same constraint. Hypothesis 3 is supported.

4.7 Discussion

The three hypotheses investigated in this chapter form a coherent account of the relationship between architectural inductive bias and representational efficiency in neural GRN inference. The cross-encoder's advantage over the two-tower (H1) is robust to realistic class imbalance where the two-tower fails (H2), and the pruning analysis explains *why* the two-tower is limited: its nominal parameter count does not reflect its effective capacity (H3).

Unified interpretation. The factorized two-tower architecture imposes a strong inductive

bias that is both its strength and its weakness. By separating TF and gene encoding, the two-tower enables efficient inference at scale: given pre-computed TF and gene embeddings, all $M \times N$ TF–gene regulatory scores can be computed as a matrix product in $O(Md + Nd)$ time rather than the $O(MNd)$ required by per-pair cross-encoder inference. This scalability property is genuine and relevant for genome-scale GRN inference over tens of thousands of gene pairs. However, the cosine-similarity scoring mechanism that enables this efficiency also prevents the model from learning the joint features that drive regulatory specificity, leading to the AUROC gap observed in H1 and the sensitivity to class imbalance in H2.

The neuron pruning results (H3) reinforce this picture: the two-tower's 512-dimensional hidden space is far larger than necessary, suggesting that the cosine similarity objective cannot leverage the representational capacity that is nominally available. This provides a mechanistic account of the H1 AUROC gap that goes beyond simply noting that the cross-encoder has a more expressive scoring function. The two-tower is not just less expressive in principle — it actively fails to use the capacity it has.

Practical recommendations. For practitioners choosing between two-tower and cross-encoder architectures for GRN inference:

- If AUROC and discriminative performance are the primary criteria, and the evaluation dataset has a realistic negative sampling ratio, the cross-encoder is the clear choice.
- If scalability is paramount — for example, in genome-scale applications requiring 10^6–10^8 pair evaluations — the two-tower may be preferred, but practitioners should be aware of the 6–10 AUROC point performance cost relative to a cross-encoder of comparable size, and should consider using only the compressed (pruned) two-tower, which achieves the same performance at 83% of the original parameter count.

Limitations. Several limitations constrain the generalizability of these findings. This study uses a single biological dataset (human brain single-nucleus RNA-seq); whether the cross-encoder advantage persists across other tissues, organisms, or expression modalities is not established. Training was performed on CPU hardware, which constrains the scale of architectures that could be practically evaluated; larger models with wider hidden layers or additional depth might close the AUROC gap. The two-tower architecture tested here uses

cosine similarity with temperature scaling; alternative scoring functions such as bilinear similarity or attention-based cross-tower interaction might recover some of the expressiveness lost to the cosine similarity bottleneck. The pruning experiment was conducted at a single seed; multi-seed pruning was not evaluated, and the non-monotonic fine-tuning behavior suggests that the 10-epoch fine-tune budget is insufficient to fully characterize the relationship between sparsity and performance. Finally, the pruning analysis characterizes only the fc1 output neurons; fc2 output neurons and the embedding tables were not pruned, and their redundancy is unknown.

Future directions. Three extensions are directly motivated by these results. First, an InfoNCE-trained two-tower, where the contrastive objective explicitly attracts positive TF–gene pairs and repels negatives within a batch, may be more robust to the marginal distribution sensitivity observed in H2 while retaining the two-tower's inference efficiency. Second, Deep Mutual Learning between two decoders may address the redundancy problem (H3) by introducing a diversity pressure through the agreement regularization term. Third, a cross-attention two-tower — where the gene encoder attends over the TF encoding before producing a final representation — would bridge the two architectures, enabling per-pair interaction features while maintaining the factorized structure required for efficient inference.

4.8 Conclusion

This chapter has presented a systematic experimental comparison of two-tower and cross-encoder neural architectures for TF–gene regulatory edge prediction from single-cell RNA-seq data. Under parameter-matched, data-matched conditions, the cross-encoder achieves 9.4 higher AUROC points than the two-tower under balanced training (0.9040 vs 0.8097), is robust to 5:1 class imbalance where the two-tower degrades by 6.6 AUROC points, and demonstrates substantially more stable training (2.6× lower standard deviation at 5:1). A neuron pruning analysis further shows that the two-tower's 512-dimensional hidden space is nearly entirely redundant: post-hoc AUROC never falls below the 95% retention threshold at any tested sparsity level, and 51 neurons per tower (10% of the original 512) suffice to match baseline performance with no additional training.

Together, these results indicate that the cross-encoder is the more appropriate choice for GRN inference tasks where discriminative accuracy is paramount, while the two-tower offers a scalable inference-efficient alternative whose performance can be maintained at substantially reduced model size through structured pruning. The mechanistic link between H1 and H3 — that the two-tower's AUROC deficit reflects wasted capacity due to cosine similarity's geometric constraints — provides a principled foundation for future work on architectures that combine the efficiency of factorized encoding with the expressiveness of joint interaction features.

4.9 References

Aibar, S., González-Blas, C. B., Moerman, T., Imrichová, H., Hulselmans, G., Rambow, F., … & Aerts, S. (2017). SCENIC: single-cell regulatory network inference and clustering. *Nature Methods*, 14(11), 1083–1086.

Bromley, J., Guyon, I., LeCun, Y., Säckinger, E., & Shah, R. (1993). Signature verification using a "Siamese" time delay neural network. *Advances in Neural Information Processing Systems*, 6.

Frankle, J., & Carlin, M. (2019). The lottery ticket hypothesis: Finding sparse, trainable neural networks. *International Conference on Learning Representations (ICLR).*

García-Alonso, L., Holland, C. H., Ibrahim, M. M., Turei, D., & Saez-Rodriguez, J. (2019). Benchmark and integration of resources for the estimation of human transcription factor activities. *Genome Research*, 29(8), 1363–1375.

Han, S., Pool, J., Tran, J., & Dally, W. (2015). Learning both weights and connections for efficient neural networks. *Advances in Neural Information Processing Systems*, 28.

Han, H., Cho, J. W., Lee, S., Yun, A., Kim, H., Bae, D., … & Lee, I. (2018). TRRUST v2: an expanded reference database of human and mouse transcriptional regulatory interactions. *Nucleic Acids Research*, 46(D1), D380–D386.

Huang, P. S., He, X., Gao, J., Deng, L., Acero, A., & Heck, L. (2013). Learning deep

structured semantic models for web search using clickthrough data. *Proceedings of the 22nd ACM International Conference on Information & Knowledge Management (CIKM)*, 2333–2338.

Huynh-Thu, V. A., Irrthum, A., Wehenkel, L., & Geurts, P. (2010). Inferring regulatory networks from expression data using tree-based methods. *PLOS ONE*, 5(9), e12776.

Kingma, D. P., & Ba, J. (2015). Adam: A method for stochastic optimization. *International Conference on Learning Representations (ICLR)*.

Nogueira, R., & Cho, K. (2019). Passage re-ranking with BERT. *arXiv preprint arXiv:1901.04085*.

Humeau, S., Shuster, K., Lachaux, M.-A., & Weston, J. (2020). Poly-encoders: Architectures and pre-training strategies for fast and accurate multi-sentence scoring. *International Conference on Learning Representations (ICLR)*.

Khattab, O., & Zaharia, M. (2020). ColBERT: Efficient and effective passage search via contextualized late interaction over BERT. *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*, 39–48.

Aibar, Sara, Carmen Bravo González-Blas, Thomas Moerman, et al. 2017. "SCENIC: Single-Cell Regulatory Network Inference and Clustering." *Nature Methods* 14 (11): 1083–86.

Bromley, Jane, Isabelle Guyon, Yann LeCun, Eduard Säckinger, and Roopak Shah. 1993. "Signature Verification Using a 'Siamese' Time Delay Neural Network." *Advances in Neural Information Processing Systems* 6.

Chen, Tianqi, and Carlos Guestrin. 2016. "XGBoost: A Scalable Tree Boosting System." *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–94.

Frankle, Jonathan, and Michael Carlin. 2019. "The Lottery Ticket Hypothesis: Finding Sparse, Trainable Neural Networks." *International Conference on Learning Representa-*

tions (ICLR).

García-Alonso, Luz, Christian H. Holland, Mahmoud M. Ibrahim, Dénes Turei, and Julio Saez-Rodriguez. 2019. "Benchmark and Integration of Resources for the Estimation of Human Transcription Factor Activities." *Genome Research* 29 (8): 1363–75.

Han, Heonjong, Jae-Won Cho, Sangyeon Lee, et al. 2018. "TRRUST V2: An Expanded Reference Database of Human and Mouse Transcriptional Regulatory Interactions." *Nucleic Acids Research* 46 (D1): D380–86.

Han, Song, Jeff Pool, John Tran, and William Dally. 2015. "Learning Both Weights and Connections for Efficient Neural Networks." *Advances in Neural Information Processing Systems* 28.

Huang, Po-Sen, Xiaodong He, Jianfeng Gao, Li Deng, Alex Acero, and Larry Heck. 2013. "Learning Deep Structured Semantic Models for Web Search Using Clickthrough Data." *Proceedings of the 22nd ACM International Conference on Information & Knowledge Management (CIKM)*, 2333–38.

Humeau, Samuel, Kurt Shuster, Marie-Anne Lachaux, and Jason Weston. 2020. "Poly-Encoders: Architectures and Pre-Training Strategies for Fast and Accurate Multi-Sentence Scoring." *International Conference on Learning Representations (ICLR).*

Huynh-Thu, Vân Anh, Alexandre Irrthum, Louis Wehenkel, and Pierre Geurts. 2010. "Inferring Regulatory Networks from Expression Data Using Tree-Based Methods." *PLOS ONE* 5 (9): e12776.

Khattab, Omar, and Matei Zaharia. 2020. "ColBERT: Efficient and Effective Passage Search via Contextualized Late Interaction over BERT." *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*, 39–48.

Kingma, Diederik P., and Jimmy Ba. 2015. "Adam: A Method for Stochastic Optimization." *International Conference on Learning Representations (ICLR).*

Nogueira, Rodrigo, and Kyunghyun Cho. 2019. "Passage Re-Ranking with BERT." *arXiv Preprint arXiv:1901.04085.*